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Lecture 22, 3/11/26

Driven, damped oscillator

EOM: $m\ddot{x} = -kx - b\dot{x} + F(t) \Rightarrow \ddot{x} + 2\beta\dot{x} + \omega_0^2 x = f(t)$

$f(t) \equiv F(t)/m$

$2\beta \equiv b/m$

$\omega_0^2 \equiv k/m$

example: $f(t) = f_0 \cos\omega t$

We changed $x \to z = x + iy$

$f_0 \cos\omega t \to f_0 e^{i\omega t}$

$\Rightarrow z(t) = Ce^{i\omega t}$ where $C = Ae^{-i\delta}$ & $A^2 = \dfrac{f_0^2}{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}$, $\tan\delta = \dfrac{2\beta\omega}{\omega_0^2-\omega^2}$

$z(t) = x(t) + iy(t)$

$\mathrm{Re}(z) = x(t) = A\cos(\omega t - \delta) \equiv x_p(t)$

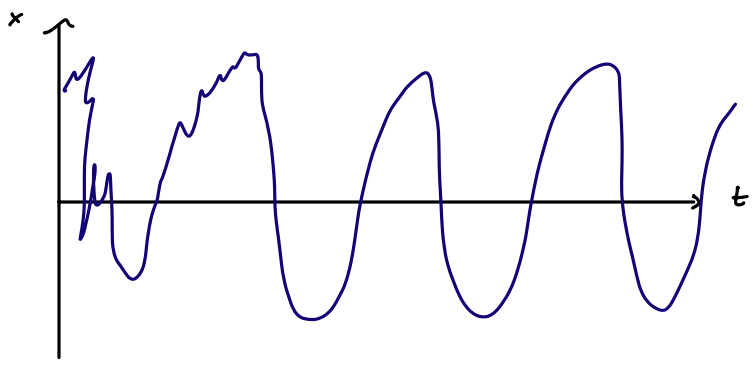
$x(t) = x_p(t) + x_H(t)$

Remember $x_H(t) = C_1 e^{r_1 t} + C_2 e^{r_2 t}$

So full solution is $x(t) = A\cos(\omega t - \delta) + C_1 e^{r_1 t} + C_2 e^{r_2 t}$

Driven, underdamped: $x(t) = A\cos(\omega t - \delta) + A_{tr}\, e^{-\beta t} \cos(\omega_1 t - \delta_{trans})$

This will lose importance over time

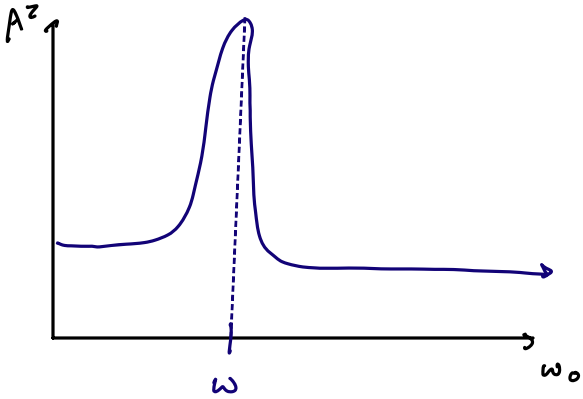


Resonance

$x(t) = A\cos(\omega t - \gamma)$

$$A^2 = \frac{f_0^2}{(\omega_0^2-\omega^2)^2 + 4\beta^2\omega^2}$$

← when denominator is small, amplitude is large



⇒ maximum is called a "resonance"

$$\text{denominator} = (\omega_0^2-\omega^2)^2 - 4\beta^2\omega^2$$

$$\frac{\partial}{\partial\omega_0}(\text{denominator}) = 2(\omega_0^2-\omega^2)(2\omega_0) = 0$$

⇒ critical points at $\omega_0 = 0$

$\omega_0 = \omega$ → minimum

$$\frac{\partial}{\partial\omega}(\text{denominator}) = 2(\omega_0^2-\omega^2)(2\omega) - 8\beta^2\omega = 0$$

$$\omega_0^2 - \omega^2 - 2\beta^2 = 0$$

⇒ critical point at $\omega = \sqrt{\omega_0^2 - 2\beta^2} \equiv \omega_2$

$$A^2\Big|_{\omega=\omega_2} = \frac{f_0^2}{[\cancel{\omega_0^2} - (\cancel{\omega_0^2} - 2\beta^2)]^2 + 4\beta^2(\omega_0^2 - 2\beta^2)} = \frac{f_0^2}{4\beta^4 + 4\beta^2\omega_0^2 - 8\beta^4} = \frac{f_0^2}{4\beta^2(\omega_0^2-\beta^2)}$$

$$A^2\Big|_{\omega=\omega_0} = \frac{f_0^2}{(\omega_0^2-\omega_0^2)^2 + 4\beta^2\omega_0^2} = \frac{f_0^2}{4\beta^2\omega_0^2}$$

$A^2\Big|_{\omega=\omega_2} > A^2\Big|_{\omega=\omega_0}$ so ω_2 maximizes A the most

If $\beta \ll 1$ then $A^2\Big|_{\omega=\omega_2} = \dfrac{f_0^2}{4\beta^2(\omega_0^2-\beta^2)} \approx \dfrac{f_0^2}{4\beta^2\omega_0^2} = A^2\Big|_{\omega=\omega_0}$

Summary of frequencies:

$\omega_0 \equiv \sqrt{k/m}$ = natural frequency = frequency for the undamped SHO

$\omega_1 = \sqrt{\omega_0^2 - \beta^2}$ = frequency of the underdamped harmonic oscillator

$\lim_{\beta \to 0} \omega_1 = \sqrt{\omega_0^2} = \omega_0$

ω = frequency of the driving force, depends on $f(t)$ for driven harmonic oscillator

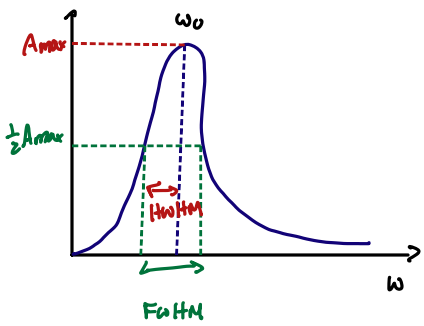
(Remember, $\beta = \frac{1}{2}\frac{b}{m}$, $\vec{F}_{drag} = -b\vec{v}$)

$\omega_2 = \sqrt{\omega_0^2 - 2\beta^2}$ = value of ω s.t. A is maximized

"resonant frequency", relevant for driven harmonic oscillator

$$A_{max} \approx \frac{f_0}{2\beta\omega_0}$$

Width of the resonance:



FWHM = full width half maximum

$\approx 2\beta$

HWHM = half width half maximum

$\approx \beta = \frac{1}{2}(2\beta) = \frac{1}{2}(\text{FWHM})$

Sharpness of peak = quality factor Q:

$$Q \equiv \frac{\omega_0}{2\beta} = \frac{\text{position of peak}}{\text{FWHM}} \Rightarrow \text{large } Q \Rightarrow \text{sharp peak}$$

$$Q = \frac{1/\beta}{2\,1/\omega_0} = \pi \frac{1/\beta}{2\pi/\omega_0} = \pi \frac{\text{decay time}}{\text{natural period}} \qquad (\tau = 2\pi/\omega_0)$$

$\Rightarrow$ large Q = long decay time

Phase: For a driven damped oscillator,

$$\delta = \tan^{-1}\left(\frac{2\beta\omega}{\omega_0^2 - \omega^2}\right) \qquad \lim_{\omega \to 0} \delta = \tan^{-1}(0) \to 0 \quad \text{for } \omega \ll \omega_0,\ \delta = 0$$

Recall $f(t) = f_0 \cos(\omega t) \Rightarrow x(t) = A\cos(\omega t - \delta) + A_{tr}\, e^{-\beta t} \cos(\omega_1 t - \delta_{tr})$

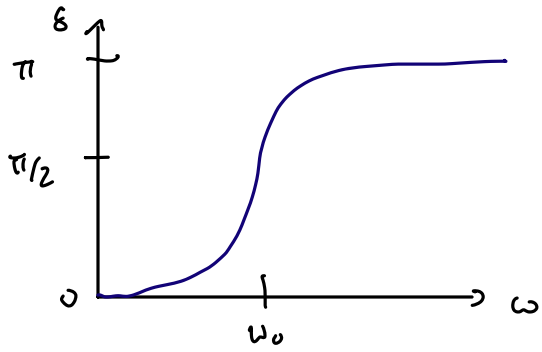
$\Rightarrow$ oscillator has nearly the same δ as driving force for small ω

OTOH, $\lim_{\omega \to \omega_0} \delta = \tan^{-1}\left(\frac{2\beta\omega_0}{0}\right) \to \pi/2$ (Remember $\tan \pi/2 = \frac{\sin \pi/2}{\cos \pi/2} = \frac{1}{0} \to \infty$)

At resonance, $x(t)$ is offset from $f(t)$ by $\pi/2$

Finally, $\lim_{\omega \to \infty} \delta = \tan^{-1}\left(\frac{2\beta}{\frac{\omega_0^2}{\omega} - \omega}\right) = \tan^{-1}\left(\frac{2\beta}{-\infty}\right) \to \tan^{-1}(-0) \to \pi$

$\tan(\pi) = \frac{\sin \pi}{\cos \pi} = \frac{0}{-1} = -0$



Fourier Series

- Fourier Series:

  Any periodic function $f(t+\tau) = f(t)$

  can be rewritten as:

$$f(t) = \sum_{n=0}^{\infty} \left[a_n \cos(n\omega t) + b_n \sin(n\omega t)\right] \qquad \omega = 2\pi/\tau$$

$$a_0 = \frac{1}{\tau} \int_{-\tau/2}^{\tau/2} f(t)\, dt \qquad [n=0]$$

$$a_n = \frac{2}{\tau} \int_{-\tau/2}^{\tau/2} f(t) \cos(n\omega t)\, dt$$

$$b_n = \frac{2}{\tau} \int_{-\tau/2}^{\tau/2} f(t) \sin(n\omega t)\, dt$$

do not just plug $f(t) = \sum_{n=0}^{\infty} a_n \cos n\omega t + b_n \sin n\omega t$ back in!

You need specific info about $f(t)$ to determine a_n, b_n

If $f(t) = \text{odd}$, $a_n = 0$

$f(t) = \text{even}$, $b_n = 0$